

Extended Data 8 | Method-headroom metrics: Baysor, Proseg, ComSeg vs oracle and nearest-nucleus (one-to-one, many-to-one homogeneity, co-assignment ARI); free-segmentation rows shaded

| regime      | method        | one-to-one | many-to-one | co-assign (ARI) | frac assigned |
|-------------|---------------|------------|-------------|-----------------|---------------|
| dense 1.43  | oracle        | 0.803      | 0.814       | 0.669           | 1.00          |
| dense 1.43  | naive         | 0.731      | 0.746       | 0.564           | 1.00          |
| dense 1.43  | Baysor        | 0.719      | 0.748       | 0.573           | 1.00          |
| dense 1.43  | ComSeg        | 0.722      | 0.737       | 0.557           | 1.00          |
| dense 1.43  | Proseg        | 0.617      | 0.628       | 0.116           | 0.83          |
| dense 1.99  | oracle        | 0.730      | 0.746       | 0.568           | 1.00          |
| dense 1.99  | naive         | 0.639      | 0.659       | 0.452           | 1.00          |
| dense 1.99  | Baysor        | 0.634      | 0.666       | 0.469           | 1.00          |
| dense 1.99  | ComSeg        | 0.635      | 0.655       | 0.448           | 1.00          |
| dense 1.99  | Proseg        | 0.564      | 0.578       | 0.093           | 0.81          |
| dense 2.63  | oracle        | 0.651      | 0.672       | 0.467           | 1.00          |
| dense 2.63  | naive         | 0.546      | 0.572       | 0.354           | 1.00          |
| dense 2.63  | Baysor        | 0.540      | 0.581       | 0.373           | 1.00          |
| dense 2.63  | ComSeg        | 0.540      | 0.568       | 0.352           | 1.00          |
| dense 2.63  | Proseg        | 0.480      | 0.500       | 0.068           | 0.80          |
| sparse 1.99 | oracle        | 0.882      | 0.888       | 0.791           | 1.00          |
| sparse 1.99 | naive         | 0.833      | 0.843       | 0.711           | 1.00          |
| sparse 1.99 | Baysor        | 0.724      | 0.749       | 0.633           | 1.00          |
| sparse 1.99 | ComSeg        | 0.768      | 0.781       | 0.630           | 0.98          |
| sparse 1.99 | Proseg        | 0.635      | 0.638       | 0.062           | 0.74          |
| repr 1.99   | oracle        | 0.808      | 0.819       | 0.677           | 1.00          |
| repr 1.99   | naive         | 0.741      | 0.756       | 0.578           | 1.00          |
| repr 1.99   | Baysor        | 0.707      | 0.735       | 0.570           | 1.00          |
| repr 1.99   | ComSeg        | 0.724      | 0.741       | 0.565           | 1.00          |
| repr 1.99   | Proseg        | 0.659      | 0.671       | 0.155           | 0.85          |
| dense 1.43  | Baysor (free) | 0.625      | 0.660       | 0.508           | 1.00          |
| dense 1.99  | Baysor (free) | 0.521      | 0.543       | 0.407           | 1.00          |
| dense 2.63  | Baysor (free) | 0.453      | 0.483       | 0.344           | 1.00          |
| sparse 1.99 | Baysor (free) | 0.712      | 0.736       | 0.619           | 1.00          |
| repr 1.99   | Baysor (free) | 0.603      | 0.616       | 0.492           | 0.99          |